5.2 The Mean Value Theorem

Name _____ Period ____ Date _____

AP Calculus 5.2 (30-minute plan) · Student sheet

I can…

- use Rolle's Theorem to find c with $f'(c) = 0$.
- find the number c guaranteed by the Mean Value Theorem.
- find where a function is increasing and decreasing.

Vocabulary · fill in the missing word

Rolle's Theorem — if f is continuous on [a, b], _____ on (a, b), and $f(a) = f(b)$, then $f'(c) = 0$ for some c in (a, b)

Mean Value Theorem — if f is continuous on [a, b] and _____ on (a, b), then $f'(c) = \frac{f(b) - f(a)}{b - a}$ for some c in (a, b)

Increasing/Decreasing Test — $f' > 0$ on an _____ means f is increasing; $f' < 0$ means f is decreasing

Part 1 · Notes and examples (fill in each note, then work the example)

Rolle's Theorem: if f is continuous on [a, b], differentiable on (a, b), and $f(a) = f(b)$, then $f'(c) =$ _____ for some c in (a, b).

Example 1

Verify Rolle's Theorem for $f(x) = x^2 - 4x$ on [0, 4] and find c.

Mean Value Theorem: under the same first two conditions, $f'(c) =$ _____ for some c in (a, b).

Example 2

Find c guaranteed by the Mean Value Theorem for $f(x) = x^2$ on [1, 3].

Increasing/Decreasing Test: if $f'(x) > 0$ on an interval, then f is _____ on that interval.

Example 3

Where is $f(x) = x^3 - 3x^2 - 9x$ increasing? Decreasing?

Part 2 · Partner practice

Partner practice

(AP® Practice) f is continuous on [1, 4] and differentiable on (1, 4), with f(1) = −2 and f(4) = 7. Which must be true?

(A) f′(c) = 3 for some c in (1, 4) (B) f′(c) = 0 for some c in (1, 4)

(C) f(c) = 8 for some c in (1, 4) (D) f′(x) > 0 for all x in (1, 4)